

Course Code: CYC01

Course Name: Engineering Chemistry

Question Paper No.: NITD/CYC01/01

Full Marks: 25

Time: 90 Minutes

Instructions:

1. Answer all questions
2. All the parts of a question must be written in same place. Otherwise, those answers will not be evaluated if written randomly.

Question No.	Body of the Question	Marks	Mappe d CO
1	(a) Explain the differences with suitable examples: (i) Ionization isomerism and Hydrate isomerism; (ii) Geometrical isomerism and Optical isomerism. (b) Calculate the CFSE value by showing the splitting under octahedral ligand field for the following ions: d^1 , d^7 and d^8	4+5= 9	CO3
2	(a) Derive the expression of work done of an n mole of ideal gas in (i) Reversible isothermal expansion, (ii) Irreversible expansion and (iii) Reversible adiabatic expansion. (b) What are the expressions of C_p and C_v in terms of enthalpy (H) and internal energy (U) respectively? Physically explain why C_p is greater than C_v . (c) Under what condition, heat change and enthalpy change will be equal?	4+3+1=8	CO4
3	(a) Give a brief discussion on the principle of IR spectroscopy. Based on this principle explain why CO_2 molecule is IR active but N_2 molecule is IR inactive. (b) How would you differentiate between the following molecules based on spectroscopic techniques? (i) cyclopentanone and pentanol (ii) 1,3-butadiene and 1,4-pentadiene	3+2+3=8	CO2

COs: CO1: Students will get the knowledge of fundamentals as well industrial applications of polymer, petroleum products, organometallic compounds and others; CO2: Students will be able to elucidate the structure of different organic compounds and to analyze the structure-property correlation; CO3: Students will be aware on the role played by different metals in biological systems and also the ecological impact of metals; CO4: Students will be able to understand and analyze Thermodynamical, kinetic as well as electrochemical aspects of chemical systems and apply the understanding in the technical field.